

Architecture

Model architectures and augmentation methods behind modern detectors

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1 AASIST: Audio Anti-Spoofing using Integrated Spectro-Temporal Graph Attention Networks

Jee-weon Jung, Hee-Soo Heo, Hemlata Tak, Hye-jin Shim, Joon Son Chung, Bong-Jin Lee, Ha-Jin Yu, Nicholas Evans

ICASSP 2022 · arXiv:2110.01200

■ <https://arxiv.org/abs/2110.01200>

ABSTRACT

Artefacts that differentiate spoofed from bona-fide utterances can reside in spectral or temporal domains. Their reliable detection usually depends upon computationally demanding ensemble systems where each subsystem is tuned to some specific artefacts. We seek to develop an efficient, single system that can detect a broad range of different spoofing attacks without score-level ensembles. We propose a novel heterogeneous stacking graph attention layer which models artefacts spanning heterogeneous temporal and spectral domains with a heterogeneous attention mechanism and a stack node. With a new max graph operation that involves a competitive mechanism and an extended readout scheme, our approach, named AASIST, outperforms the current state-of-the-art by 20% relative. Even a lightweight variant, AASIST-L, with only 85K parameters, outperforms all competing systems.

2 End-to-End Anti-Spoofing with RawNet2

Hemlata Tak, Jose Patino, Massimiliano Todisco, Andreas Nautsch, Nicholas Evans, Anthony Larcher

ICASSP 2021 · arXiv:2011.01108

■ <https://arxiv.org/abs/2011.01108>

ABSTRACT

Spoofing countermeasures aim to protect automatic speaker verification systems from attempts to manipulate their reliability with the use of spoofed speech signals. While results from the most recent ASVspoof 2019 evaluation show great potential to detect most forms of attack, some continue to evade detection. This paper reports the first application of RawNet2 to anti-spoofing. RawNet2 ingests raw audio and has potential to learn cues that are not detectable using more traditional countermeasure solutions. We describe modifications made to the original RawNet2 architecture so that it can be applied to anti-spoofing. For A17 attacks, our RawNet2 systems results are the second-best reported, while the fusion of RawNet2 and baseline countermeasures gives the second-best results reported for the full ASVspoof 2019 logical access condition. Our results are reproducible with open source software.

3 Automatic Speaker Verification Spoofing and Deepfake Detection Using Wav2vec 2.0 and Data Augmentation

Hemlata Tak, Massimiliano Todisco, Xin Wang, Jee-weon Jung, Junichi Yamagishi, Nicholas Evans
Odyssey 2022 (Speaker and Language Recognition Workshop) · arXiv:2202.12233
■ <https://arxiv.org/abs/2202.12233>

ABSTRACT

The performance of spoofing countermeasure systems depends fundamentally upon the use of sufficiently representative training data. With this usually being limited, current solutions typically lack generalisation to attacks encountered in the wild. Strategies to improve reliability in the face of uncontrolled, unpredictable attacks are hence needed. We report in this paper our efforts to use self-supervised learning in the form of a wav2vec 2.0 front-end with fine tuning. Despite initial base representations being learned using only bona fide data and no spoofed data, we obtain the lowest equal error rates reported in the literature for both the ASVspoof 2021 Logical Access and Deepfake databases. When combined with data augmentation, these results correspond to an improvement of almost 90% relative to our baseline system.

4 RawBoost: A Raw Data Boosting and Augmentation Method applied to Automatic Speaker Verification Anti-Spoofing

Hemlata Tak, Madhu Kamble, Jose Patino, Massimiliano Todisco, Nicholas Evans
ICASSP 2022 · arXiv:2111.04433
■ <https://arxiv.org/abs/2111.04433>

ABSTRACT

This paper introduces RawBoost, a data boosting and augmentation method for the design of more reliable spoofing detection solutions which operate directly upon raw waveform inputs. While RawBoost requires no additional data sources, e.g. noise recordings or impulse responses and is data, application and model agnostic, it is designed for telephony scenarios. Based upon the combination of linear and non-linear convolutive noise, impulsive signal-dependent additive noise and stationary signal-independent additive noise, RawBoost models nuisance variability stemming from, e.g., encoding, transmission, microphones and amplifiers, and both linear and non-linear distortion. Experiments performed using the ASVspoof 2021 logical access database show that RawBoost improves the performance of a state-of-the-art raw end-to-end baseline system by 27% relative and is only outperformed by solutions that either depend on external data or that require additional intervention at the model level.

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